Date: | **Class:** | **Topic:**
xx-yy-zzzz | Science 101 | Some more examples for supported markdown structures [Letter format]

💡 Cues / Questions

You can create/modify your own header styles too.

Top and bottom heights are limited (lower limits are 5mm):

- Top height limit: 80mm ~ 3.15″
  - usually used as header
- Bottom height limit: 100mm ~ 3.94″
  - usually used as summary

Chemistry Formulas:

$\mathrm{Fe^{II}Fe^{III}{}_2O_4}$

$\mathrm{A} \xrightarrow{\mathrm{H_2O}} \mathrm{B}$

$\mathrm{A{-}B{=}C{\equiv}D}$ chem. bonding

$C_p[\mathrm{H_2O\,(l)}] = 75.3\ \frac{\mathrm{J}}{\mathrm{mol\,K}}$

Math formula sizing by using standard LaTeX syntax:

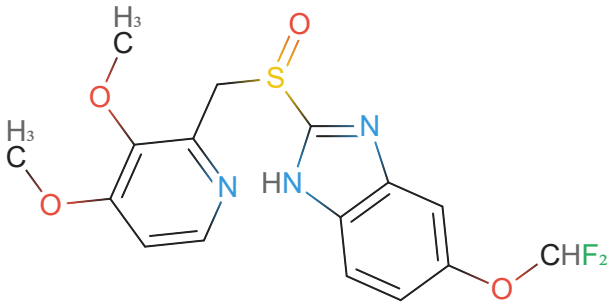
1. `\tiny` – ca. **5–6 pt**
2. `\scriptsize` – ca. **7 pt**
3. `\small` – ca. **9 pt**
4. `\normalsize` – **10–12 pt** (not needed!)
5. `\large` – ca. **12–14 pt**
6. `\Large` – ca. **14–16 pt**
7. `\LARGE` – ca. **17–20 pt**
8. `\huge` – ca. **20–24 pt**
9. `\Huge` – ca. **25–30 pt** in MathJax.

📖 Notes / Details

Chemical Reaction Equations with mhchem Package: mhchem Syntax

$$\mathrm{Zn^{2+}} \underset{+2\,\mathrm{H^+}}{\overset{+2\,\mathrm{OH^-}}{\rightleftharpoons}} \underset{\text{amphoteres Hydroxid}}{\mathrm{Zn(OH)_2}\downarrow} \underset{+2\,\mathrm{H^+}}{\overset{+2\,\mathrm{OH^-}}{\rightleftharpoons}} \underset{\text{Hydroxozikat}}{[\mathrm{Zn(OH)_4}]^{2-}}$$

Molecules with SMILES:



The molecular formula of Pantoprazole: $\mathrm{C_{16}H_{15}F_2N_3O_4S}$ is created with LaTeX code. The molecule picture was made using SMILES notation code.
To get codes for both use a search engine with terms: "Pantoprazole SMILES" and "Pantoprazole mhchem LaTeX". It's much faster and more accurate then to create it yourself. Copy & paste - it's done! For not so well known molecules ask an LLM to generate it.

A hint for beginners in bioinformatics, chemistry, biology and medicine:
Counting atoms in molecular formula and comparing numbers within SMILES pics and code you might ask: Why are atoms like hydrogen (H) or carbon (C) often missing? Here's why: SMILES simplifies molecular structures by omitting atoms. **In short, "missing atoms" in SMILES code from chemical databases and pictures are not errors – they are simplifications based on well-defined chemical rules and conventions**. These apply to other atoms too: nitrogen, oxygen, sulfur, halogens and even metals as well. All depends on conventions used in chemical databases or software interpreting the notation. Without support, knowing these conventions would be crucial for accurately interpreting or creating SMILES strings. But getting the code online from databases, AI (LLM's) or search engines is 99% reliable. So even as an early stage student, middle schooler or interested newbie you might not be familiar with those rules, but you can still get your beautiful Cornell notes done.

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5. large-size:

$$\frac{\partial u}{\partial t} = p^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

6. Large-size:

$$\frac{\partial u}{\partial t} = p^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

7. LARGE-size:

$$\frac{\partial u}{\partial t} = p^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

8. huge-size:

$$\frac{\partial u}{\partial t} = p^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

9. Huge-size:

$$\frac{\partial u}{\partial t} = p^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

1. Tiny:

$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = (ad - bc)$

2. Script-size:

$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = (ad - bc)$

3. Small-size:

$\sum_{i=1}^{n} i^2 = \frac{1}{2}n(n+1)$

4. Standard-size:

$\sum_{i=1}^{n} i^2 = \frac{1}{2}n(n+1)$

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet. Lorem ipsum dolor sit amet, consetetur sadipscing elitr

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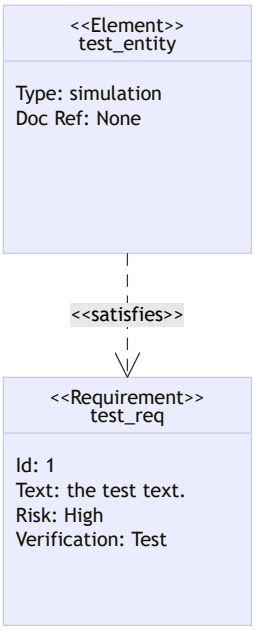
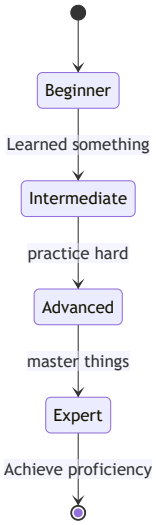
xx-yy-zzzz	Science 101	Some more examples for supported markdown structures [Letter format]

Tag-Pills for highlighting keywords:

#chemistry #biology #math #topics





My custom callout heading

- Lorem ipsum dolor sit
  - consetetur sadipscing elitr

- 1 custom type (choose Title and color with hex) syntax: **> [!custom|#1a20d9] My custom callout...**
- All 12 basic callout types (not the original Obsidian callout boxes, so with limited functionality)

Note
invidunt ut labore et dolore aliquyam erat

Abstract
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Info
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Failure
invidunt ut labore et dolore aliquyam erat

Tip
invidunt ut labore et dolore aliquyam erat

Danger
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Success
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Bug
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Question
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Example
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Warning
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Quote
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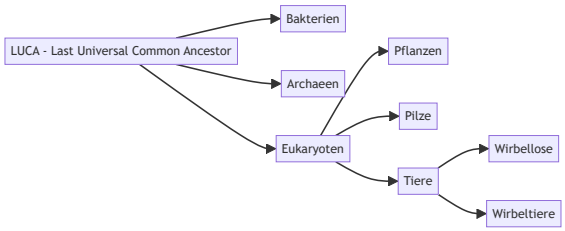
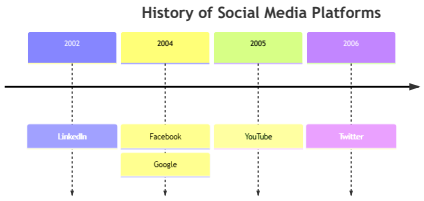
Some Mermaid Diagramms:

The syntax is quite easy to learn, but there are also graphical tools available and of course LLM's. Several software tools offer a **graphical user interface (GUI)** for creating Mermaid diagrams and exporting them as Mermaid code: **Mermaid Flow**, **Mermaid Chart**, **Miro** and **Eraser**